

OFUNATO, Japan

WMO: 475120

Lat: 39.0650N

Lon: 141.7150E

Elev: 126

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.9	25.0	8.1	8.4	28.6	10.3	9.3	29.7	21.7	34.6	19.4	33.9	5.1	340	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.4	87.0	74.6	83.8	73.6	81.3	72.6	76.3	83.3	75.3	81.4	74.2	79.6	7.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.4	129.2	79.2	73.3	124.5	78.6	72.3	120.0	77.8	39.9	83.8	38.9	81.7	37.9	79.6	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.8	13.9	DB	19.4	92.8	2.6	2.9	17.6	94.9	16.0	96.6	14.6	98.2	12.7	100.3	(4)
(5)			WB	17.4	78.1	2.4	1.2	15.7	79.0	14.3	79.7	12.9	80.4	11.2	81.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	33.8	34.6	39.9	49.1	58.0	64.4	71.9	74.7	68.6	58.3	47.6	38.4
	DBStd	15.05	4.28	4.74	5.37	6.11	5.48	4.85	5.29	4.65	5.24	5.26	6.00	5.08
	HDD50	1827	503	431	317	90	6	0	0	0	0	4	116	360
	HDD65	4958	968	851	778	479	225	67	6	1	22	215	522	824
	CDD50	3063	0	0	4	62	254	433	680	765	559	261	44	1
	CDD65	720	0	0	0	1	8	50	221	301	131	8	0	0
	CDH74	3668	0	0	0	8	55	167	1150	1772	502	14	0	0
	CDH80	852	0	0	0	2	9	27	292	455	67	0	0	0
Wind	WSAvg	5.7	6.4	6.3	6.4	6.4	6.0	5.4	4.8	4.7	4.8	5.1	5.3	6.1
Precipitation	PrecAvg	58.7	2.0	1.8	3.5	4.7	5.7	6.5	7.7	6.9	7.8	5.6	3.9	2.4
	PrecMax	77.6	6.6	5.4	6.9	10.9	11.5	12.2	19.1	23.9	19.8	14.1	13.2	8.0
	PrecMin	39.6	0.4	0.3	0.6	0.6	1.8	1.9	2.0	1.1	1.1	1.7	0.3	0.4
	PrecStd	8.8	1.7	1.3	1.8	2.6	2.5	2.5	4.1	5.2	4.0	2.8	3.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.2	54.4	63.1	73.2	80.5	83.8	91.0	92.0	85.5	75.6	66.5	57.8
		MCWB	43.2	45.3	49.6	54.9	60.6	69.8	75.1	76.5	73.8	65.8	56.8	51.5
	2%	DB	46.5	49.5	56.9	67.0	74.3	78.4	86.5	87.8	81.7	71.9	63.4	53.9
		MCWB	40.0	42.1	47.6	53.2	59.4	66.7	74.2	75.8	72.5	63.5	55.4	47.7
	5%	DB	43.7	45.8	53.0	63.1	70.6	75.0	83.1	84.6	79.0	69.4	60.9	50.4
		MCWB	38.0	39.0	44.9	51.3	57.9	65.2	73.2	74.8	71.4	62.3	54.0	44.8
	10%	DB	41.2	42.8	49.8	59.3	67.3	72.1	80.0	82.1	76.7	67.3	58.1	47.4
		MCWB	35.7	36.5	42.8	50.0	57.2	64.3	72.1	73.9	70.4	60.4	51.8	41.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.8	47.2	52.6	59.4	65.4	71.8	77.0	78.3	76.3	69.3	60.5	53.7
		MCDB	47.9	51.7	59.2	67.8	72.8	79.1	85.3	86.3	81.6	72.0	63.3	56.5
	2%	WB	41.3	43.2	49.1	55.8	62.6	69.1	75.5	76.9	74.6	66.5	57.6	48.9
		MCDB	45.2	48.3	54.7	62.5	68.3	75.2	82.8	84.4	79.0	70.2	61.6	52.6
	5%	WB	38.7	39.9	46.4	53.9	60.8	67.2	74.4	75.9	73.3	63.7	55.1	45.3
		MCDB	43.1	44.8	51.7	59.5	66.8	72.3	80.9	82.5	77.6	67.6	59.4	49.9
	10%	WB	36.3	37.1	43.7	51.9	59.1	65.7	73.1	74.9	71.6	61.5	52.7	42.3
		MCDB	40.7	41.9	49.3	57.5	65.1	70.4	78.8	80.8	75.8	65.8	57.1	46.7
Mean Daily Temperature Range	5% DB	MDBR	11.5	12.9	14.7	15.9	14.9	12.2	11.2	11.4	12.4	14.7	15.1	12.4
		MCDBR	14.9	17.4	20.0	21.5	20.6	17.0	16.0	15.1	13.9	15.4	18.3	17.1
		MCWBR	11.2	12.3	13.7	12.0	10.1	8.1	6.8	6.4	7.1	9.6	12.5	13.3
	5% WB	MCDBR	14.6	17.0	18.3	17.6	16.4	13.6	14.1	13.2	11.7	12.6	16.8	16.7
		MCWBR	12.0	13.2	14.5	11.4	9.8	7.2	6.6	6.1	6.9	9.6	13.2	13.9
Clear-Sky Solar Irradiance	taub	0.363	0.409	0.488	0.534	0.554	0.538	0.544	0.526	0.466	0.430	0.397	0.372	
	taud	2.171	2.037	1.886	1.836	1.851	1.971	2.028	2.076	2.208	2.195	2.187	2.182	
	Ebn at Noon	246	250	241	239	238	241	239	239	247	241	232	233	
	Edh at Noon	34	44	57	64	65	57	54	50	42	38	33	31	
All-Sky Solar Radiation	RadAvg	669	923	1207	1483	1599	1557	1405	1392	1158	947	695	562	
	RadStd	40	55	65	134	152	102	192	178	106	93	44	43	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page